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Single-source chip-based frequency comb enabling extreme parallel data transmission

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The Internet today transmits hundreds of terabits per second, consumes 9% of all electricity worldwide and grows by 20–30% per year[1,2]. To support capacity demand, massively parallel communication links are installed, not scaling favourably concerning energy consumption. A single frequency comb source may substitute many parallel lasers and improve system energy-efficiency[3,4]. We present a frequency comb realized by a non-resonant aluminium-gallium-arsenide-on-insulator (AlGaAsOI) nanowaveguide with 66% pump-to-comb conversion efficiency, which is significantly higher than state-of-the-art resonant comb sources. This enables unprecedented high data-rate transmission for chip-based sources, demonstrated using a single-mode 30-core fibre. We show that our frequency comb can carry 661 Tbit s^{-1} of data, equivalent to more than the total Internet traffic today. The comb is obtained by seeding the AlGaAsOI chip with 10-GHz picosecond pulses at a low pump power (85 mW), and this scheme is robust to temperature changes, is energy efficient and facilitates future integration with on-chip lasers or amplifiers[5,6].

As worldwide data traffic continues to grow, it becomes urgent to utilize communication resources optimally, facilitating capacity growth at reduced energy consumption. Current optical communications systems, based on wavelength-division multiplexing (WDM) and single-core single-mode fibre, are reaching a capacity limit due to nonlinearities in fibre[7]. To ensure capacity growth, space-division multiplexing (SDM) using multicore fibres or multimode fibres has been proposed[8–15] in addition to WDM, where each core or each mode carries parallel WDM channels. This massive parallelization approach in turn increases the demand for even more power-hungry arrays of discrete wavelength laser sources[14]. To address the issue of continuously installing more parallel laser sources, optical frequency combs have become interesting, as they are generated using a single laser as the seed, and one may replace large numbers of parallel lasers by the individual comb lines, resulting in much lower energy consumption and smaller space occupation[11]. Recently, various chip-based frequency comb sources have been demonstrated for WDM data transmission, such as 12 Tbit s^{-1} transmission based on a mode-locked laser[16]. However, mode-locked lasers tend to have a limited bandwidth. Frequency combs have also been demonstrated by soliton comb generation in microresonators and even used for up to 50 Tbit s^{-1} data transmission[17]. This technique is challenged by a low pump-to-comb conversion efficiency, with often only <1% of the pump being converted to the generated frequency comb in the bright soliton comb case[17]. Dark soliton comb generation has recently been shown to exhibit a conversion efficiency up to 20% within a limited bandwidth allowing for 4 Tbit s^{-1} transmission, limited by the reliance on waveguides with normal dispersion[18]. In addition, microresonators inherently rely on a delicate balance between Kerr nonlinearity, dispersion and thermal resonance shifts to generate soliton frequency combs[19]. This introduces a need for active stabilization to ensure long-term stability[20].

In contrast, broadband frequency combs can also be achieved by spectral broadening of a narrow-band frequency comb[3,4], which exhibits a high pump-to-comb conversion efficiency. Furthermore, the non-resonant nature of the scheme makes it more resilient to temperature changes and facilitates long-term stability. Frequency comb broadening based on highly nonlinear fibres (HNLFs) has been used for optical communications[3,4,15]. Typically, tens to hundreds of metres of HNLF is required to achieve good conversion, which renders HNLFs too bulky for integration on a chip. Comb broadening has also been demonstrated in integrated waveguides of various material platforms[21,22], with the use of femtosecond pump pulses with high peak powers and typically with low repetition rates (<1 GHz). Since the pulse repetition rate is equal to the frequency spacing of the comb lines and determines the highest baud rate of a data signal, pump pulses with high repetition rates between 10 GHz and 50 GHz are desired for optical communications. The peak-to-average power ratio of such pulses is much lower than that of femtosecond pulses at low repetition rates, making it very challenging to realize efficient on-chip frequency comb broadening at high rates.

Here, we explore chip-based frequency combs as single-source emitters for optical communications, and in particular investigate how much data can be carried on the light from a single chip. We encode de-correlated data on 2,400 parallel SDM/WDM channels with 16-state quadrature amplitude modulation (16-QAM) in combination with Nyquist optical time-division multiplexing (TDM) and polarization-division multiplexing (PDM). In this way, six physical dimensions (amplitude, phase, time, frequency, polarization and space) are used for data modulation and multiplexing to boost the total data rate. We demonstrate the transmission of 661 Tbit s^{-1} data carried on the broadened frequency comb from a single highly nonlinear AlGaAsOI nanowaveguide[23,24], which is pumped by 10-GHz picosecond pulses with an average launched

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power of only 85 mW. The non-resonant AlGaAsOI waveguide is operated without a temperature controller or a feedback loop for stabilization. The low loss and high nonlinearity characteristics of the AlGaAsOI waveguide enables efficient frequency comb broadening, and the high pump-to-comb conversion efficiency allows for sufficient power per comb line at the chip output.

AlGaAsOI has recently emerged as an ultra-efficient nonlinear platform, combining a high intrinsic material nonlinearity with a large refractive index contrast between the nonlinear medium and the cladding materials[25]. The bandgap of AlGaAs can be engineered by changing the Al concentration to avoid two-photon absorption at telecom wavelengths[26]. All of this renders AlGaAsOI a good platform for self-phase modulation (SPM)-based optical frequency comb spectral broadening (Fig. 1a). Owing to the large index contrast of this structure, light can be strongly confined in a submicrometre waveguide core (Fig. 1b), which enhances the light–matter interaction and results in an ultra-high nonlinear parameter (γ) of ~660 W^{-1} m^{-1} for an AlGaAsOI nanowaveguide[23]. Compared with a typical HNLF, the AlGaAsOI nanowaveguide exhibits three orders of magnitude higher figure of merit, defined as the ratio of the maximal achievable nonlinear phase shift and the nonlinear medium length (Supplementary Information). In addition, the waveguide dispersion dominates over the material dispersion for subwavelength-sized waveguides, and hence the group velocity dispersion can be engineered to be anomalous at telecom wavelengths (Supplementary Fig. 1), which is desired to achieve efficient SPM-induced frequency comb broadening.

The frequency spacing between the comb lines is determined by the repetition rate of the seed pulses (Fig. 2), which is locked to the 10-GHz clock from the transmitter and can be controlled with high accuracy at Hz level (Supplementary Fig. 5). The generated frequencies can be stably aligned to the ITU (International Telecommunication Union) grid by fine tuning the seed laser centre frequency[27]. The on-chip frequency comb broadening exhibits a very high pump-to-comb conversion efficiency of 66%, thus offering a more power-efficient solution than the soliton comb generation based on resonant enhancement[19]. The laser linewidth is another key property of a frequency comb, as a narrow linewidth is required to be able to modulate both the amplitude and phase of the optical signal. The linewidths of the broadened frequency comb at the wavelengths of 1,529.97 nm and 1,560.09 nm are measured to be 40 kHz at both wavelengths (Supplementary Fig. 2), which are the same as that of the seed mode-locked laser and well below the requirement for the 16-QAM modulation at 10 GBaud (ref. [28]). Thus, we did not observe linewidth degradation due to the comb broadening within the measured wavelength range, as expected given the correlated phase of the seed laser[29].

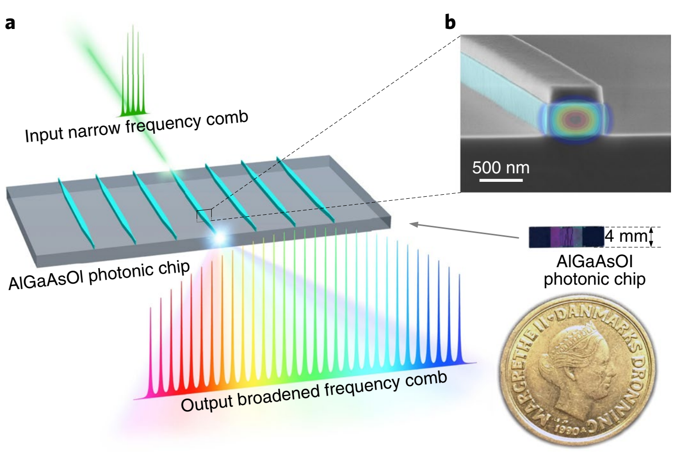

Fig. 1 | AlGaAsOI photonic chip. a, A narrow-band frequency comb is spectrally broadened to a broadband frequency comb through self-phase modulation by passing through an AlGaAsOI nanowaveguide. **b**, The AlGaAsOI photonic chip is smaller than a coin, but can accommodate hundreds of nanowaveguides. The scanning electron microscopy image shows the cross-section of the nanowaveguide with cross-section dimensions of 280 nm × 600 nm, indicating strong light confinement (the simulated field distribution for the fundamental transverse electric (TE) mode has been artificially added).

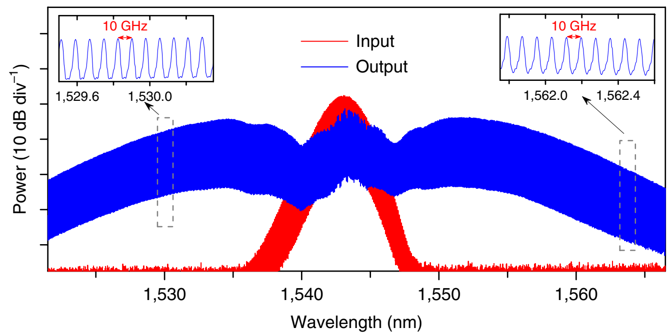

Fig. 2 | Frequency comb broadening in an AlGaAsOI nanowaveguide. A 10-GHz, 1.5-ps pulse generated from a mode-locked laser is launched into the AlGaAsOI nanowaveguide, with an average launched power of only 85 mW. The input frequency comb (20-dB bandwidth of 6.4 nm) is spectrally broadened by SPM to a 20-dB bandwidth of ~44 nm, covering more than the telecom C band. All comb lines are equidistantly spaced by 10 GHz, exactly following the repetition rate of the seed laser. Insets: zoom-in of the broadened frequency comb.

To investigate how much data the generated frequency comb can sustain, it is tested in a challenging scenario of multidimensional modulation and multiplexing. Figure 3a shows the concept of six-dimensional data modulation and multiplexing for large-capacity optical data transmission using a frequency comb source. Amplitude and phase are used for data modulation; time, polarization, wavelength and space are used for multiplexing, that is, as independent dimensions where each adds additional data channels. The generated comb lines serve as the WDM source for multiple wavelength channels. Figure 3b–d show the scheme of the large-capacity six-dimensional data transmission using a single-source chip-based frequency comb. The broadened frequency comb at the output of the AlGaAsOI photonic chip is amplitude equalized using a wavelength-selective switch (WSS)[30] and then split into 30 copies. Each of the frequency comb copies is further separated in wavelength using WDM demultiplexers and act as light sources for 80 WDM channels. Each WDM channel is modulated by 10 GBaud 16-QAM, encoding 4 bits on each pulse, multiplexed in time by a factor of K ($K = 4$ in this case), and multiplexed by orthogonal polarizations using a polarization beam combiner (PBC). The data rate of each modulated TDM–PDM–16-QAM channel is 10 Gbit s^{-1} × 4 (16-QAM modulation) × 4 (TDM) × 2 (PDM) = 320 Gbit s^{-1}. These channels are multiplexed in wavelength (×80) by a WDM multiplexer and finally launched into the 30-core fibre through a fan-in device, resulting in 2,400 parallel data channels with an aggregate gross data rate of 768 Tbit s^{-1} (320 Gbit s^{-1} × 2,400).

The heterogeneous single-mode 30-core fibre uses a trench-assisted refractive index structure and an intercore phase mismatching by the use of four types of core to reduce mode coupling between adjacent cores[12], as shown in Supplementary Fig. 10. As a result, a high-density core arrangement with intercore crosstalk below −50 dB after 9.6 km is achieved. The low crosstalk of the 30-core fibre facilitates low-complexity reception of the SDM channels without computational-intensive multiple-input-multiple-output

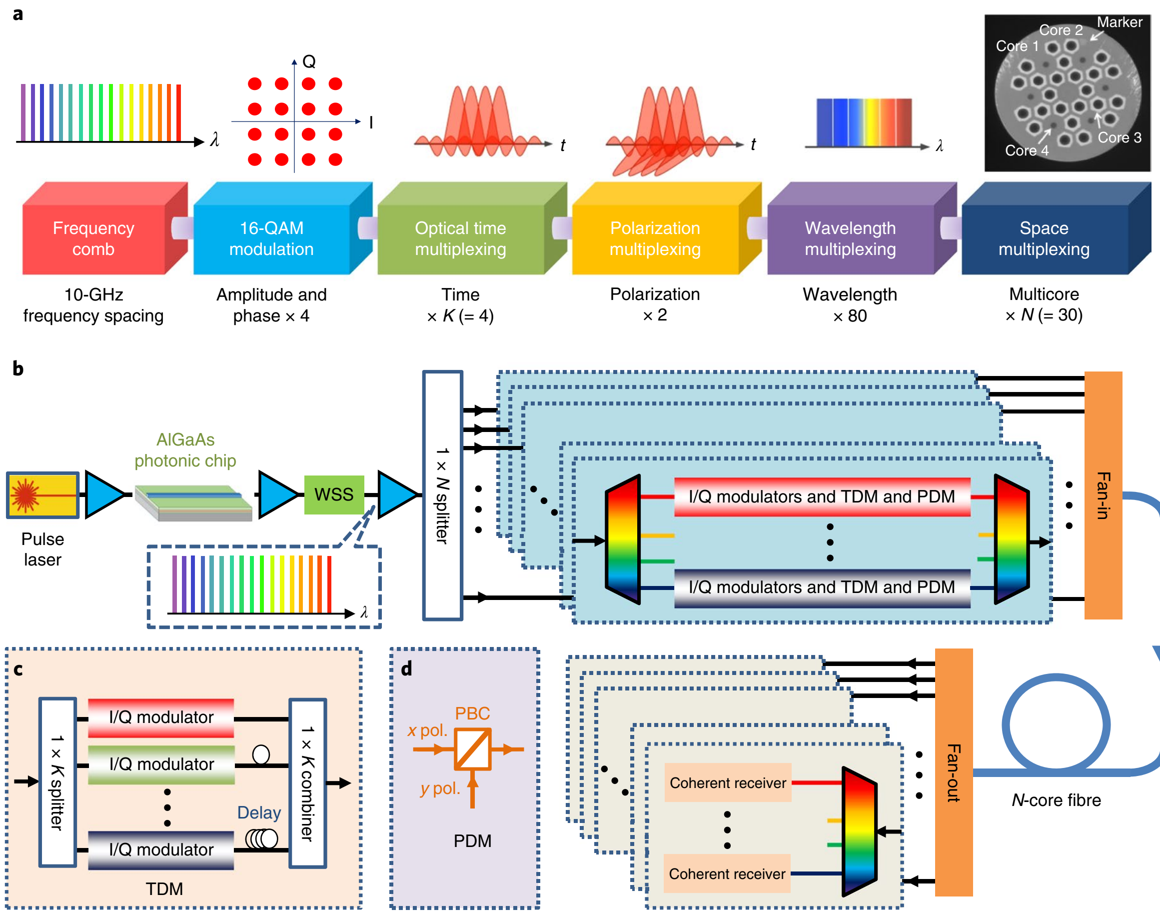

Fig. 3 | Generation and transmission of multi-100 Tbit s^{-1} data carried by the AlGaAsOI SPM-based frequency comb. a, Concept of frequency comb source based on six-dimensional data modulation and multiplexing. The chip-based frequency comb is used as WDM sources. Amplitude and phase are modulated with 16 states in the complex plane of the optical field, corresponding to four bits of information per pulse. Optical pulses are temporally interleaved and then filtered in the frequency domain using a rectangular spectral shape, resulting in sinc-shaped waveforms in the time domain. Data are multiplexed in two orthogonal polarizations. Eighty WDM channels are multiplexed and launched into a multicore fibre with N cores ($N=30$ in this case). **b**, Implementation scheme of large-capacity six-dimensional data transmission using chip-based frequency comb source. **c**, Time-division multiplexing (TDM) based on time-domain pulse interleaving in a passive fibre-delay multiplexer. **d**, Polarization-division multiplexing (PDM) by combining two orthogonal polarizations (x pol. and y pol.) using a polarization beam combiner (PBC).

(MIMO) processing. The 30 cores are arranged within a cladding diameter of 228 μm, which is less than the 250 μm constraint to maintain mechanical stability comparable to that of standard single-mode fibres.

The schematic of the implemented experimental set-up is shown in Fig. 4a. The single-source laser in the transmitter is a mode-locked laser, which generates 10-GHz pulses with a full-width at half-maximum (FWHM) of 1.5 ps. The pulses are amplified and launched into the 5-mm-long AlGaAsOI nanowaveguide, with an average launched power of only 85 mW (peak power of ~5.6 W), to achieve SPM-based frequency comb broadening. The average power of the broadened frequency comb at the output of the chip is ~13 mW. This corresponds to an average power of ~32 μW (or −15 decibel relative to 1 mW (dBm)) per comb line for ~400 comb lines within the telecom C band. Note that to achieve a desirable optical signal-to-noise ratio (OSNR) after data transmission, the comb lines should have a power level of around 40 dB above the quantum noise limit (around −61 dBm for a single polarization in 0.1-nm bandwidth)[18], that is, 8 μW (or −21 dBm). The central part (5 nm) of the frequency comb has a large power variation, resulting in OSNR degradation for the low-power part[2]. Therefore, the central part from 1,540.73 nm to 1,545.87 nm is replaced with the original spectrum from the mode-locked laser by passing through the other path of the first WSS, resulting in a flat and high-quality frequency comb (Fig. 4b). More details about the experimental set-up are given in the Methods.

To properly evaluate the chip-based single source for carrying multi-100 Tbit s^{-1}, all comb lines are simultaneously data modulated and then launched into the 30-core fibre (Fig. 4c). Using this approach, we treat all the 2,400 SDM/WDM channels alike and we sequentially characterize each of them. To evaluate the quality of the transmitted data, bit error rates (BER) are measured after transmission for all the 80 WDM channels over the 30 spatial channels (Fig. 4d). For the data channels with a BER below a certain forward error correction (FEC) limit, a BER $< 10^{-15}$ can be achieved after the error correction[31], which is considered error-free performance. All the 80 WDM channels are below FEC limits, with 58 WDM channels below the soft-decision FEC (SD-FEC) limit (20% overhead) and 22 WDM channels below the hard-decision FEC (HD-FEC) limit (7% overhead). Therefore, the BER measurements confirm the successful transmission of all the 2,400 channels with a net rate of 661 Tbit s^{-1} after FEC overhead subtraction. This achievement is mainly enabled by the high nonlinearity and low linear and nonlinear loss of the AlGaAsOI nanowaveguide, which offers sufficient

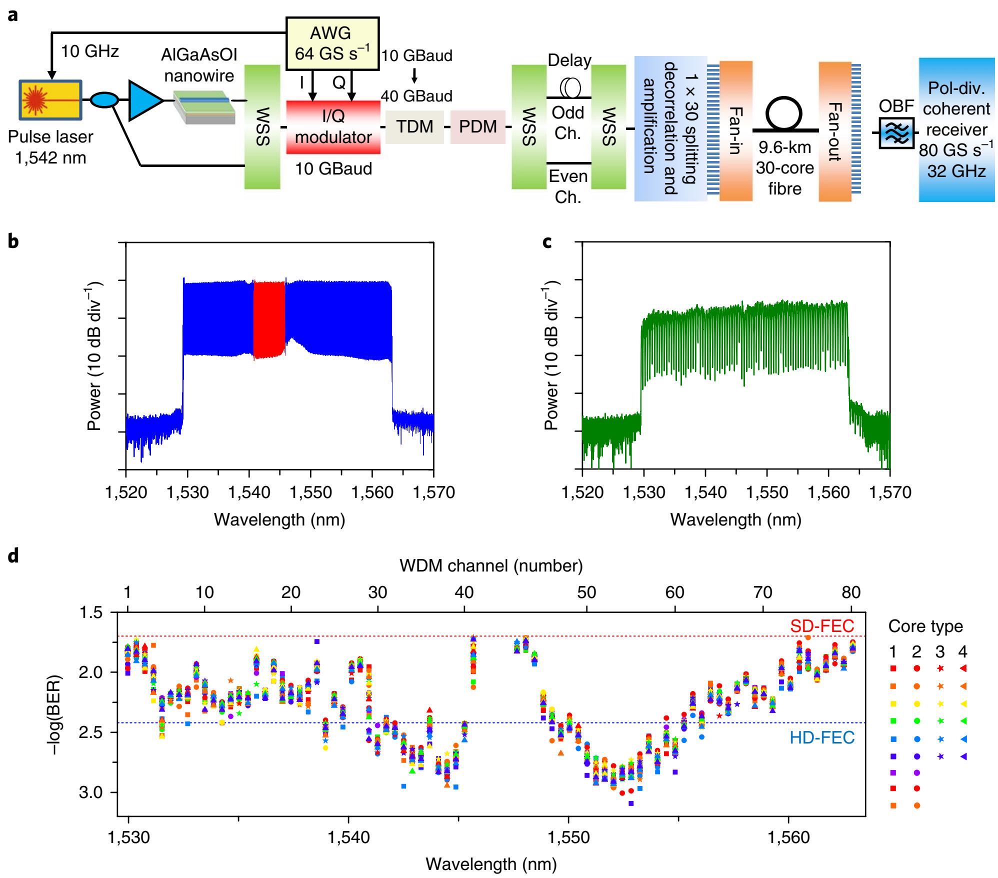

Fig. 4 | 661 Tbit s^-1 data transmission using chip-based frequency comb source. **a**, Schematic of the implemented experimental set-up for the transmission of 661 Tbit s^{-1} with 2,400 parallel channels over a 30-core fibre. OBF, optical bandpass filter. GS s^{-1}, GSamples s^{-1}. **b**, The generated frequency comb is amplitude equalized (blue) in a wavelength-selective switch and the low-quality part in the centre is replaced by the original spectrum (red) from the seed laser, resulting in a flat and high-quality frequency comb, which is subsequently data modulated and transmitted. **c**, The comb spectrum with data after 9.6-km 30-core fibre transmission and amplification. **d**, BER measurements for the 80 WDM channels over the 30 spatial channels. All the 2,400 WDM/SDM channels are below the FEC limits (58 WDM channels below the SD-FEC limit (BER = 2×10^{-2}), red line; 22 WDM channels below the HD-FEC limit (BER = 3.8×10^{-3}), blue line), which can achieve a BER < 10^{-15} after the error correction. A few centre channels have BERs above the FEC limits due to degraded OSNR, not shown in the figure.

frequency comb output power with a low pump power. The measured BERs of all channels depend on wavelength, resulting from the spectral shape of the broadened frequency comb and the increased noise at the edge of the C-band.

To evaluate the power efficiency of the chip-based frequency comb, we compare it with parallel WDM lasers. The power savings of using the AlGaAsOI chip-based frequency comb source scales with the number of WDM channels. The power consumption of using the single-source frequency comb is only 1/4 to 1/15 of the power consumption of using parallel WDM lasers for 80 to 400 channels (number of lasers) (Supplementary Information).

We have presented a photonic chip-based frequency comb with a sufficient comb output power to support several hundred Tbit s^{-1} optical data generation and transmission. This is the highest data rate using a chip-based frequency comb source, which may potentially replace hundreds of parallel lasers with reduced power consumption. With further optimization of the dispersion of the AlGaAsOI waveguide, the shape of the broadened frequency comb could become even smoother and the OSNR of the low-power part of the frequency comb could be increased. This may allow for employing even higher order modulation formats and achieve higher transmission rates. With the low pump power requirement, this scheme has potential for realization of a fully integrated frequency comb source including chip-based pump lasers[5] and amplifiers[6]. Other components required for equalization and splitting, such as multichannel variable optical attenuator arrays and WDM (de-)multiplexers, also have integration solutions[32,33]. Moreover, the comb source is scalable to even broader bandwidths such as S, C and L bands (Supplementary Information), and is thus promising for photonic chip-based single-source transmission beyond Pbit s^{-1}.

Methods
Methods, including statements of data availability and any associated accession codes and references, are available at https://doi.org/10.1038/s41566-018-0205-5.

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Author contributions

H.H. conceived and designed the experiments. F.D.R. and E.P.d.S. conducted digital signal processing for the transmitted data. M.P. designed the AlGaAs device. H.H., F.D.R., F.Y., K.I. and M.N performed the transmission experiment. H.H., F.D.R and M.P. analysed the data. M.P., L.O., E.S. and K.Y. fabricated the AlGaAs device. H.H. and M.P. characterized the AlGaAs device. F.Y., Y.A., Y.S. and T.Mi. designed the multicore fibre. Y.A. and Y.S. fabricated the multicore fibre. T.Mi., Y.M., P.G., D.Z., M.G., L.K.O. and T.Mo. contributed to the experiment. H.H. and L.K.O. wrote the manuscript and all the co-authors contributed to the writing. Y.M., K.Y., T.Mo. and L.K.O. supervised the projects.

Competing interests

The authors declare no competing interests.

Additional information

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Methods

Design and fabrication of the AlGaAsOI nanowaveguides. We have developed an AlGaAsOI platform[23], where a thin $Al_xGa_{1-x}As$ layer on top of a low-index insulator layer resides on a semiconductor substrate as shown in Fig. 1b. Wafer bonding and substrate removal are used to realize AlGaAsOI wafers[25]. Electron-beam lithography (JEOL JBX-9500FS) was used to define the nanowaveguide pattern in the electron-beam resist hydrogen silsesquioxane (HSQ, Dow Corning FOX-15). The nanowaveguide pattern was then transferred into the AlGaAs layer using a boron trichloride (BCl_3)-based dry etching process in an inductively coupled plasma reactive ion etching (ICP-RIE) machine. As the refractive index of HSQ is relatively low (similar to SiO_2), it was kept on top of the AlGaAs device pattern. Finally, the device pattern was clad in a 3-μm-thick silica layer using plasma-enhanced chemical vapour deposition (PECVD). Owing to the large index contrast (~55%) between AlGaAs and silica, light can be confined in the submicrometre waveguide core. The strong light confinement not only effectively enhances the device nonlinearity but also enables efficient dispersion engineering. The performance of nanowaveguide devices is also limited by the linear loss induced by light scattering due to surface roughness. To reduce the surface roughness, all the fabrication processes were optimized. High-quality epitaxial material growth and substrate removal are required to ensure smoothness for top and bottom waveguide surfaces, while high-quality electron-beam lithography and dry-etching processes ensure a small roughness on the waveguide sidewall surfaces (Fig. 1b). The propagation loss of the fabricated AlGaAsOI waveguide is about 1.5 dB cm^{-1} (ref. [23]). The chip was cleaved to form the input and output facets where nanotapers enabled efficient chip-to-fibre coupling for characterization. Tapered fibres are used at both facets and the coupling loss is ~3 dB per facet.

Data modulation and multiplexing. In our experiment, the broadened frequency comb with 10-GHz spacing is modulated with 10 GBaud 16-QAM in a standard I/Q modulator driven by a 60 GSample s^{-1} arbitrary waveform generator (AWG) with 20 GHz of analog bandwidth. The modulated 10-GBaud 16-QAM signal is optical time-division multiplexed (OTDM) by a factor of four to 40 GBaud based on time-domain pulse interleaving in a passive fibre-delay multiplexer (MUX × 4). A delay-and-add polarization multiplexing emulator is used to generate PDM signals, corresponding to the data rate of 320 Gbit s^{-1} per wavelength channel (40-GBaud serial rates, 4 bits per symbol by 16-QAM modulation and two polarizations). In addition, optical Nyquist filtering was used to strongly confine the spectrum of each WDM channel, resulting in a rectangular spectrum in the frequency domain and sinc-function waveforms in the time domain with the duty cycle of 25% (Fig. 3a). Although the sinc-shaped pulses strongly overlap with neighbouring pulses, intersymbol interference (ISI) is minimized by having one sinc-pulse's zero-crossing coincide with its neighbour's peak.

To generate a WDM signal with 50-GHz spacing, the broadened frequency comb is spectrally sliced into odd and even channels and separated into two paths using a second WSS. The delay difference between the two paths is 7.5 ns in order to de-correlate the odd and even channels. This is a laboratory emulation of a real WDM transmission system, which can be used to characterize the transmission performance[34]. All the WDM channels are modulated by one I/Q modulator before the splitting and de-correlation in the lab experiment. The I/Q modulator has a maximum input power of 32 mW (that is, 15 dBm) and an insertion loss of ~17 dB. The insertion loss is mainly limited by operating the I/Q modulator close to the linear regime. Therefore, the average input power for each of ~400 comb lines is limited to be −11 dBm and the average output power for each comb line is only −28 dBm at the output of the modulator. The modulated WDM channels are amplified by an erbium-doped fibre amplifier (EDFA). The signal OSNR will be significantly degraded with the low power at the input of the EDFA. Since the average power (−28 dBm) for each comb line at the output of the modulator is much lower than the average power (−15 dBm) for each comb line at the output of the AlGaAs chip, the signal OSNR is mainly limited by the noise generated in the EDFA at the output of the modulator. In real WDM systems, the WDM channels are modulated independently with parallel modulators, and the input power of the modulator for each WDM channel is significantly higher. Therefore, the real-world transmission performance is underestimated in the lab experiment, limited by available lab equipment, and better transmission performance is expected for real WDM systems.

The second WSS in the experimental set-up is programmed for rectangular filtering with a bandwidth of 40 GHz and 50-GHz spacing (compatible with the ITU standard), in order to generate 40 GBaud Nyquist TDM–PDM–16-QAM signals for all the WDM channels. A 10-GHz guard band was inserted in between the WDM channels to minimize interchannel crosstalk. The odd and even WDM channels are recombined using a third WSS. As a result, 80 WDM channels between 1,529.97 nm and 1,562.92 nm are generated to reach a data rate of 25.6 Tbit s^{-1} (Fig. 4c). Thirty SDM channels are fully de-correlated to each other, with at least 2.5 ns between SDM channels. These SDM channels are multiplexed by a 3D-waveguide-based fan-in device, and then launched into the 9.6-km heterogeneous single-mode 30-core fibre. The launched power for each core is between 14–18 dBm accounting for loss variations in the fan-in/fan-out from 5 dB to 8 dB.

Coherent receiver. After the 30-core transmission, the 30 spatial channels are demultiplexed using another 3D-waveguide-based fan-out device. A tunable bandpass filter with a bandwidth of 50 GHz is used to select each WDM channel. The selected WDM channel is detected with a dual-polarization coherent receiver, consisting of a polarization-diversity 90-degree hybrid, a local oscillator (LO) and four balanced detectors, followed by a digital sampling oscilloscope (80 GSamples s^{-1}, 33-GHz bandwidth). The waveforms were processed from 0.5 million samples by offline digital signal processing (DSP) including adaptive time-domain equalization using constant and multi-modulus algorithms, OTDM demultiplexing, carrier recovery by decision-directed phase-locked loop, demapping and BER counting. Carrier recovery, demapping and BER counting were performed on each OTDM tributary separately.

Data availability. The data that support the plots within this paper and other findings of this study are available from the corresponding author upon reasonable request.

References

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